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DISPATCHER (DISPAT,104) 10/2/73 JGC/BPC/JBL

/CHAN 16 BREAK

```
DISPAT,  LIF"U"SCF"U"CLL
        DIO DSP1          /SAVE FLAGS
        RCL 2S           /TOP OF IO CLEAR
        EEM
        SAD ([DDTS6+3]"T"4)      /IS IDDT "EXECUTING" INSTR?
        JMP DSP3          /YES, GO ADJUST PC FOR CALLING SEQUENCES
DSP4,    RTB
        DIO TRPBUF
        CLA"U"SWP
        SMA              /AN IOT?
        JMP ELGIOT       /NO
        RAL 5S          /YES, TEST I BIT
        SPA
        JMP DSP16       /YES, CORE 16 IOT
        RAL 1S
        DCH (JMP 140000 DSP2)    /REGULAR DISPATCH
        RCL 6S
        LFI
DSP2,    JMP I DSPTB+0
DSP3,    LAW I 1         /DDT PC ADJUST
        ADD I (DDTS6+4)
        DAP 71
        JMP DSP4
DSP16,   RAL 1S         /CORE 16 DISPATCH
        DCH (JMP 140000 DSP5)
        RCL 6S         /GET REDISPATCH BITS INTO IO
        LFI
        LAC 70         /MOVE AC,PC,IO
        DAC I (C16AC)
        LAC 71
        DAC I (C16PC)
        LAC 72
        DAC I (C16IO)
        LAC DSP1       /FLAGS, TOO
        DAC I (C16FLA)
        LAC I (BITS)
        AND (-40000)   /RESTART MODE DEBREAK LEGAL BIT
        DAC I (BITS)
DSP5,    LAC C16TB+0    /GET DEBREAK DISPATCH
DSP12,   DAC 71        /DEBREAK, DISPATCH
        JMP I 71
DSP1,    0             /FLAGS
TRPBUF,  0
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ILGIOT,  EEM
ELGIOT,  LIO I (BITS)  /ILLEGAL INSTR.
          LAI
          AND (-500000) /IDDT AND RESTART BITS
          DAC I (BITS)
          SPI I        /IS GUY UNDER IDDT
          JMP DSP8      /NO
          LAC C16BKP    /BREAKPOINT, GET DDT
DSP9,    LIO 71        /SET UP CRASH AC,PC,IO,C16 TEMP STORAGE
          DAC 71
          DIO I (HH16PC)
          JSP C16TSV
          LIO 70
          DIO I (HH16AC)
          LIO 72
          DIO I (HH16IO)
          LIO DSP1
          DIO I (HH16FG)
          LIO TRPBUF
          JMP I 71

DSP8,    RIL 1S
          SPI
          JMP XDDBK      /HELD BY XDDBK + CRASHED
          LAC TRPBUF
          SAS (100400)   /HLT?
          JMP DSP10
          RIL 3S
          SPI
          JMP REALHALT
          LAC (20000)    /REALHALT LEGAL BIT
          IOR I (BITS)
          DAC I (BITS)
          LAC C16HLT
          JMP DSP12

DSP10,   LAC RCORE      /REMEMBER PHYSICAL CORE,..
          DAP TRPBUF    /...FOR HH TYPEOUT ETC
          LAC C16HH     /HALT HORRIBLE
          JMP DSP9

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/HELD BY XDDT + CRASHED; TYPE ON SOROBAN

```

XDDTBK,  LAW I STAT
          ADD RSTAT
          RCR 9S
          LAW I 3
          LSM
          JDA 14SOCT
          LAW 77
          AND I (TTNO)
          RCR 6S
          LAW I 2
          JDA 14SOCT
          LAC (DCH XDDTBT)
          JDA 14STXT
          LAC (CORHNG)
          DIP I ERSTAT
HANG1E,  ESM
          JMP HANG1

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XDDTBT,  276464
          237756          /XDDT[CR][EOM]

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REALHALT, LEM
           LAC (UNUSED)
           DAC I RSTAT
           LAW WHERE-STAT
           ADD RSTAT
           DAP REALH1
           LAW 300
REALH1,   DAC .
           DZM I RCORE
REALH3,   LAW I 1
           LSM
           ADD WTOP
           DAP REALH2
           LIA
           LAW 300
REALH2,   SAS .
           JMP HANG1E
           DIO WTOP
           ESM
           JMP REALH3

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/DEBREAK FROM CORE 16 THRU HH16
C16RH,   LAW 77
          AND I (BILLTT)
          JDA TTNUM
          LAW C16RH1
          DAP TTCKSX
          JMP TTCKS1+2

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C16RH1,  SAS (2)
          JMP C16RH2
          LAC C16BKK
          DAC 71
          JMP R0

C16RH2,  LAC (400000)
          IOR I (BITS)
          DAC I (BITS)
          LIO I (HH16AC)
          DIO 70
          LIO I (HH16PC)
          DIO 71
          LIO I (HH16IO)
          DIO 72
          LIO I (HH16FG)
          JMP R0+1

HANG,    RCR 6S           /HANG GUY WITH STATUS IN AC (12-17)
HANG+1,  JSP HANGS       /SUBROUTINE IS IN TT ROUTINES
HANG1,   ISB 1700
          DZM HOTFLG     /TELL SWAPPER
          JMP R0

.RCK,    RCK             /REAL RCK
          JMP DPEEK1

FF8,     SZF I 5
          JMP FF88
          SZF 6
          IOT+743        /LIGHT UP THE SLEEPY OPERATOR
          SZF I 6
          IOT+343
          JMP R1

FF88,    LAC 70           / 8 FLIP=FLOPS
          DCH (ADD DCH .+1)
          IOT+70         /0X=CLEAR ALL, 1X=SET X, 2X=CLEAR X
          JMP R1

DELAY,   IDX 71          /HANG FOR SECONDS (ALSO CALLED FROM CORE 15)
          LAW 70
DELAY2,  DAP DELAY3      /SEE DRPA,DPPA ALSO
          LAC I (BILLTT)
          SCR 9S
          SZA I           /SKIP IF A QUEUE ALREADY SAVED UP
          LAC I ERSTAT
          RCL 9S
          DAC I (BILLTT)
          LAW 7777
DELAY3,  SUB .
          DAP I ERSTAT
          LAW CLKHNG"T"100
          JMP HANG

```

/RESTART MODE CONROL IOT

```
RSMC,      SZF I 5
            JMP RSMC1      /DON'T DEBREAK
            LAC (40000)    /DEBREAK LEGAL BIT
            AND I (BITS)
            SZA I
            JMP ELGIOT
            LAW I 1
            ADD I (RESTPC)
            DAC 71
            LAC (170000
            AND I (RESTPC
            LIA
            SAD (160000
            CLA
            SZA
            JMP ELGIOT
            JSP C16TUS      /UNSAVE CORE 16 IF NECESSARY
```

```
RSMC1,     LAC I (BITS)
            AND (-140000)  /RESTART MODE AND DEBREAK LEGAL BITS
            SZF 6
            IOR (100000)   /RESTART MODE BIT
            DAC I (BITS)
            JMP R1
```

/DISPATCH CLEAR THE READER

```
DCLR,      JSP RDINIT
            JMP R1          /OWNED BY ANOTHER USER
            JMP R2          /OK
```

/DISPATCHER RPA

```
DRPA,      LAW (1)
            SZS 60
            JMP DELAY2
            JSP RDA
            JMP ELGIOT      /UNOWNED BY USER
            JMP . 3
            DIO 72
            JMP R1          /RETURN TO USER WITH CHAR IN IO
            LAW RPAHNG"T"100
            JMP HANG
```

/DISPATCHER RELEASE THE READER

```
DRLSRD,    JSP RDRLS
            JMP R1
```

/GET READER ROUTINE

```

XRDIN,  LIF"U"SCF
RDINIT,  DAC RX
          LAC RDWHO
          SPA
          /OR SAD (-0)
          LAC RSTAT
          DAC RDWHO
          SAS RSTAT
          JMP RDIN1
          SZF 1
          JMP RDIN2
          LAC RBO
          DAC RBI
          LAC CRB1
          DAC RB1
RDIN2,   IDX RX
RDIN1,   LFI
          JMP I RX

```

/GET A CHARACTER FROM THE READER BUFFER

/R1=NOT YOURS, R2=EMPTY, R3=OK, CHAR IN IO

```

RDA,     DAC RX
          LAC RSTAT
          SAS RDWHO
          JMP I RX
          IDX RX
          LAC RB1
          SAD RB2
          JMP R4
          /READER RUNNING
          LAC RBI
          SUB RBO
          RAL 1S
          SPA
          ADD (RDLEN RDLEN)
          SUB (20)
          SMA
          /START READER WHEN 20 CHARS IN BUFFER
          JMP R4
          LAC RB2
          DAC RB1
RB2,     RPA
          RCK
          DIO RBK
R4,      LAC RBO
          SAD RBI
          JMP I RX
          /EMPTY
          LIO I RBO
          ADD (400000)
          SAD RBE
          SUB (RDLEN)
          DAC RBO
          SPA
          RIR 9S
          RCR 8S
          CLI
          RCL 8S
          / IO BITS 10-17
          IDX RX
          JMP I RX

```

```
RDRLS,    DAC RX          /RELEASE THE READER
          CLI"U"CM1
          LAC RSTAT
          SAD RDWHO
          DIO RDWHO
          JMP I RX
```

RX, 0

```
/DISPATCHER GET THE PUNCH
DGETPU,   LAC PUNWHO
          SPA              /OR SAD (-0)
          LAC RSTAT
          DAC PUNWHO
          SAS RSTAT
          JMP R1
          JMP R2
```

```
/DISPATCHER RELEASE THE PUNCH
DRLSPU,   CLI"U"CM1
          LAC RSTAT
          SAD PUNWHO
          DIO PUNWHO
          JMP R1
```

```
/DISPATCHER PPA
DPPA,     LAC RSTAT
          SAS PUNWHO
          JMP ELGIOT
          LAW (1)
          SZS 60
          JMP DELAY2
          LAC PUP
          IDA
          SAD PEND
          SUB (PUNLEN)
          LIO (PPAHNG
          SAD PSP
          JMP HANG+1
          LIO 72
          DIO I PUP
          DAC PUP
          CLA
          SAS PRUN
          ISB 1200
          JMP R1
```

/IOT TO START A USER

PSUC,	ESM
PSU,	LEM
	LAW WHERE=1
	DAP PSUA
PSUD,	IDX PSUA
	SAD (SAS WHERE+100)
	JMP HANG1
	LAW 300
PSUA,	SAS .
	JMP PSUD
	LIO (500)
	LSM
	SAS I PSUA
	JMP PSUC
	DIO I PSUA
	LAW 7777
	AND PSUA
	SAD WTOP
	IDX WTOP
	ESM
	LAW I WHERE=STAT
	ADD PSUA
	DAP PSUB
	LAW 1
PSUB,	DAC .
	DZM HOTFLG
	JMP R1

/WRITE P POINTER

```

DWPP,      LAC (=JMP)
            AND 70
            SUB (11)
            SAS (=10)      /OK TO CHANGE JOB NUMBER
            SZS 50
            JMP DWPP1
            SPA
            JMP R1
            SUB (I=11)
            SAD (PRIORITY=I)      /CHANGING HIS PRIORITY?
            JMP DWPP2      /YES
            SMA

```

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DWPP1,     JMP R1
            LAC 72
            DAC I 70
            JMP R1

```

```

DWPP2,     LAW I 7777      /SET NEW PRIORITY
            AND 72
            LIA
            DCH (LAC PRIORITY)
            SPI
            JMP R1      /NEG. PRIORITY CANNOT BE THE HIGHEST REACHED
            SUB I (BITS) /COMPARE WITH PREVIOUS HIGH
            XOR I (BITS)
            RAR 7S
            SWP
            SPI I
            DCH (JMP BITS) /SAVE NEW HIGHEST PRIORITY
            JMP R1

```

/READ P-POINTER

```

DPEEK,     LIO I 70
            SZF 1
            LIO I 72
DPEEK1,    SZF 2
            DIO 72
            SZF I 2
            DIO 70
            JMP R1

```

```

/SUPER-PEEK AND SUPER-WPP,  EXAMINE OR CHANGE OTHER USER CORES,
SPP,      LAC SPOWN
          SZA      /LOGIC ALREADY IN USE?
          JMP SP2   /YES
          LAW SPHANG
          DAP SPX   /SET SWITCH TO HANG OWNER
          LAC 70    /USER AND CAD BEING EXAMINED
          DAP SPCAD
          AND (770000
          RAL 6S
          ADD (STAT
          DAC SPUSER /STAT PTR TO USER TO BE EXAMINED
          SAD BDSTAT /UNHANG HIM
          LAW FSTAT
          DAP SP1
          LAW I 7777
          LEM
          LSM
SP1,      AND .
          SAD (UNUSED
          JMP SP3    /USER DOES NOT EXIST
          AND (300000
          CLI"U"SWP
          SNI I
          DIP I SP1
          LAW 777
          AND I SP1  /MASK QUEUE IN CASE CLKHNG
          DAP I SP1
          ESM
          EEM
          LAW 7777
          AND I NETHND
          SAD RSTAT
          JMP SP4    /NETHANDLER DOESN'T NEED SS5
          SZS I 50
          CLF 1
SP4,      DZM SPFLAG /SET READ-WRITE FLAG, ZERO IF READING.
          SZF 1
          IDX SPFLAG
          LAW I 1
          DAC SPCNT
          LAC 72    /CONTENTS TO BE WRITTEN
          DAC SPWORD
          SZF 2     /ONE WORD OR MANY WORDS ?
          JMP SP5
SPHANG,   LAC RSTAT
          DAC SPOWN
          LIO (SPPHNG
          SAD SPUSER
          JMP HANG1  /LOOKING AT SELF, DON'T HANG.
          JMP HANG+1

```

SP2,	SAD RSTAT	
SPX,	JMP .	/SWITCH. SPHANG OR SPDONE.
	LAW 7777	/TOO BAD, LOGIC IS BUSY.
	AND I NETHND	
	SAD RSTAT	/IS THE "NCP" TRYING TO GET THE LOGIC ?
	JMP SP6	
	LAC SPOWN	
SP3,	ESM	
	DAC 70	/ERROR CODE. STAT PTR OR "UNUSED" STATUS.
	JMP R1	/ERROR RETURN
SP6,	LAC SPOWN	/TOO BAD FOR NORMAL USER.
	SAD BDSTAT	/THE "NCP" IS USURPING THE LOGIC.
	LAW FSTAT	
	IOR (DCH)	
	DAC T1	
	LAC (300000)	
	LSM	
	AND I T1	
	CLI"U"SWP	
	SNI I	
	DIP I T1	/UNHANG FORMER OWNER
	LAW 777	
	AND I T1	
	DAP I T1	
	ESM	
	DZM SPOWN	
	DZM SPUSER	
	JMP SPP	
SP5,	DZM T1	
	DAP T1	
	RAL 6S	
	CMA	
	IOR (777740)	
	DAC T3	
	DAC SPCNT	
	SZF I 1	
	JMP SPHANG	
	LAW I 40	
	ADD SPCAD	
	SPA	
	JMP ILGIOT	
	LAW SPWORD	
	DAP .+2	
	LAC I T1	
	DAC .	
	IDX .-1	
	IDX T1	
	ISP T3	
	JMP .-5	
	JMP SPHANG	

SPDONE,	DZM SPOWN	/RELEASE SPP LOGIC
	SZF 1	
	JMP R2	/GOOD RETURN AFTER WRITING INTO OTHER USER
	SZF 2	
	JMP .+4	
	LAC SPWORD	/CURRENT CONTENTS
	DAC 70	
	JMP R2	/GOOD RETURN, ONE WORD READ
	LAW 7777	
	AND 72	
	DAC T2	
	SUB (40)	
	SPA	
	JMP ILGIOT	
	LAW SPWORD	
	DAP .+1	
	LAC .	
	DAC I T2	
	IDX .-2	
	IDX T2	
	ISP SPCNT	
	JMP .-5	
	JMP R2	/GOOD RETURN, MANY WORD READ
SPOWN,	0	/0 OR STAT PTR TO OWNER
SPUSER,	0	/STAT PTR TO USER BEING EXAMINED
SPCAD,	0	/CAD WITHIN CORE OF SPUSER
SPWORD,	.-40/	/CONTENTS
SPFLAG,	0	/0 FOR "READ", NON-ZERO FOR "WRITE".
SPCNT,	0	/NUMBER OF WORDS TO TRANSFER
T1,	0	/THREE TEMPORARIES FOR DISPATCHER OR SWAPPER
T2,	0	
T3,	0	

/DISPATCHER TT ROUTINES

/SET UP PTRS USED BY DISPATCHER TT ROUTINES

```

TTSET,    DAP TTSETX
          LAW 37          /ONLY 32. TTYS THESE DAYS
          AND I (TTNO
          JDA TTNUM
TTSETX,    JMP .

TTNUM,     0              /JDA WITH TT NUMBER IN AC
          DAP TTNUMX
          LAW TTP"1"200000
          ADD TTNUM
          SAL 29          /GET PLACE IN TTP TABLE
          DAP TTFP
          IDA
          DAP TTSP
          IDA
          DAP TTUP
          DAP TTDCHU
          DAP TTLCHU
          IDA
          DAP TTPP
TTNUMX,    JMP .

TTFP,      140000+.      /FLAGS
TTSP,      140000+.      /SERVICE PTR
TTUP,      140000+.      /USER PTR
TTPP,      140000+.      /PROGRAM

```

/CHECK TT STATUS IOT

```

TTCKS,     JSP TTCKS1
          SZA I
          JMP TTRETN
          DAC 70          /PUT ERROR CODE IN AC AND ERROR CODE WORD
          JMP TTCKS2      /CLEAR OUT BREAK IF ANY

```

/SUBROUTINE TO CHECK OWNERSHIP AND BREAK

```

TTCKS1,    DAP TTCKSX
          JSP TTSET      /TT NUMBER IS IN USER CORE
TTCKS1+2,  LAW 377
          AND I TTPP
          LIO (1
          SAS RSTAT
          JMP TTCKSY      /NOT HIS. ERROR CODE 1.
          LAW I 7767      /770010 TO AC
          AND I TTFP
          RAR 49
          RIL 19
          SPQ            /BREAK?
          CLI            /NO
TTCKSY,    LAI
TTCKSX,    JMP .

```

/SUBROUTINE NORMALLY USED BY TT IOT'S. CHECKS OWNERSHIP
 /AND DETECTS "BREAK" STATUS AND CLEARS IT. GIVES TRAP (OR R1)
 /UNLESS EVERYTHING OK.

```

TTOK,      DAP TTOKX
           JSP TTCKS1
           SZA I
TTOKX,     JMP .
TTCKS2,    SAS (2
           JMP TTERR      /NOT HIS TT
DHANGX,    LAC I ERSTAT    /BREAK, GET USER'S STATUS
           RAL 3S
           SPA             /"BREAK" BIT SET?
           JMP R0          /BREAK OCCURRED SINCE CH 16 BREAK BEGAN
           LAW 10          /CLEAR OUT BREAK STATUS
           LSM
           AND I TTFF
           SZA I
           DIP I TTFF      /CLEAR NULL COUNT UNLESS BEING INTERRUPTED
           ESM
           LAW 2           /ERROR CODE FOR BREAK
TTERR,     DAC I (ERCODE)  /SAVE ERROR CODE
           LIO 71          /SAVE ADDRESS OF ERROR IN TRAPPC
           DIO I (TRAPPC)
           SZF 6
           JMP R1
           LAW TTTSU
           SPI
           IOR (400000)    /SET OVERFLOW IF PREVIOUSLY ON
           DAC 71
           JMP R0
  
```

/IOT TYI. TYPE IN CHAR TO TOP 6 BITS OF AC.

```

TYI=.      JSP TTOK
           JSP TTU        /GET CHAR OR HANG
           DAC 70         /STORE CHAR
           JMP TTRETN
  
```

/IOT TYO. TYPE CHAR IN TOP 6 BITS OF AC.

```

TYO=.      JSP TTOK
           LAW I 7777
           AND 70
           JDA TTS        /TYPE CHAR OR HANG
TTRETN,    SZF 6          /TEST WHETHER 1= OR 2=RETURN IOT
  
```

```

R2,        IDX 71        /NORMAL RETURN 2
R1,        IDX 71        /NORMAL RETURN 1
R0,        LIO DSP1      /NORMAL RETURN 0
R0+1,      LFI          /RESTORE FLAGS
           LAC 70
           LIO 72
           JMP I 71
  
```

```

/IOT TIS, TYPE IN STRING TERMINATED BY 74, AC IS PTR,
TIS, JSP TTOK
      LAC I (TISMAX
      IDC
      DAC DTEM1
TIS1, LAC 70 /LOOP, TEST FOR VALID PTR,
      IDC"U"SCI
      SAD DTEM1
      JMP TIS2 /TISMAX EQUALLED, WILL BE EXCEEDED,
      AND (177740
      RCL 6S
      SNI"U"SZ A I
      JMP TIS3 /TOO LOW OR TOO HIGH
      JSP TTU /GET CHAR OR HANG
      DCH I 70
      SAD (74
      JMP TTRETN /TERMINATOR SEEN
      JMP TIS1

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TIS2, LAW 3 /TISMAX ERROR CODE
      JMP TTERR

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TIS3, LAW 4 /ILLEGAL REGION ERROR CODE
      JMP TTERR

```

```

/IOT TOS, TYPE STRING TERMINATED BY 74, AC IS PTR,
TOS, JSP TTOK
TOS1, LIO 70
      LCH I 70 /GET CHAR
      SAD (740000
      JMP TTRETN /RETURN, LEAVING AC STEPPED.
      DIO 70 /UNSTEP IN CASE OF HANG
      JDA TTS /TYPE CHAR OR HANG
      LCH I 70 /STEP AC
      JMP TOS1

```

```

/SUBROUTINE TO TAKE 6-BIT CHAR OUT OF BUFFER, HANG USER IF EMPTY.
TTU, DAP TTIX /EXIT THROUGH TTS
      ERG

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```

      LAW I 2
      DSC 600
      AND I TTFP
      SAS I TTFP
      JMP TTUFUL /PF 5 SET, BUFFER FULL WITH PTRS EQUAL.
      RCR 6S /SAVE PF 1 IN I.O.
      LAC I TTUP
      SPI I /TYPEACTIVE?
      SAD I TTSP /EMPTY?
      JMP TIHANG /TYPEACTIVE OR BUFFER EMPTY
TTU1, IDC /STEP USER PTR
      DAC I TTUP
      JMP TTSS

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TTUFUL, DAC I TTFP /CLEVERLY CLEARS PF 5
        LAC I TTUP
        JMP TTU1

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```

/SUBROUTINE TO TYPE 6 BITS IN TOP OF AC.  HANG USER IF BUFFER FULL.
TTS,      0
          DAP TTSX
          LAW I 7777
          SAD TTS
          JMP TTS7      /WARNING CHAR.  GOBBLE IT UP.
          ERG
          LAW 40
          LIO I TTUP
          DSC 600
          IOR I TTFP    /STF 1
          SAS I TTFP    /ALREADY SET?
          JMP TTSBEG    /NOT TYPEACTIVE.  START HIM UP.

TTS1,     LAI
          IDC
          SAD I TTSP
          JMP TOHANG    /BUFFER FULL
          LIO I TTPP
          RIL 8S
          SPI I
          JMP TTS2      /77 NOT SAVED UP
          IDC
          SAD I TTSP
          JMP TOHANG    /NO ROOM FOR 2-BYTE CHAR
          CLC
          XCT TTDCHU    /STORE 77
TTS2,     LAC TTS
          DCH I .        /STORE CHAR
TTS3,     LAW I 1000
          AND I TTPP
TTS4,     DAC I TTPP    /SET OR CLEAR "SAVED 77" STATUS
TTS5,     JSP ASC6      /RE-ACTIVATE CH 6
TTLCHU,   LCH .        /GET CHAR FOR TTU RETURN
          LRG
TTSX,     JMP .

TTSBEG,   DIO I TTSP    /SET PTRS EQUAL
          AND (-2       /CLF 5
          DAC I TTFP
          AND (211      /KEEP LINK, PF 3, AND PF 6
          SAS (1
          JMP TTS1      /EXPECTING A BREAK: DON'T TCB
          LIO TTNUM
          SSB
          LAW TTTBL"T"10000 /CALCULATE PLACE IN TBL
          IOR TTS
          RAL 6S
          LIO I TTPP
          RIL 8S
          SPI           /"SAVED 77" KEPT IN "1000" BIT
          ADD (100      /77 SAVED UP

```



```

DAP . 1
LIO .           /GET CODE
TCB             /SEND IT
SAS (TTTBL+115
SAD (TTTBL+100
JMP TTS6        /BREAK OR CR=NO-LF, SET LINK,
JMP TTS3

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TTS6,    LAW 200           /SET LINK
IOR I TTFF
DAC I TTFF
JMP TTS3

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TTS7,    LAW 1000         /SET "SAVED 77" STATUS
IOR I TTTP
JMP TTS4

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```

/ROUTINES TO HANG USER, ETC.
TIHANG,  LIO (TYIHNG
JMP TTHANG

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```

TOHANG,  LIO (TYOHNG
TTHANG,  LRG
          IDX TTSX         /IN CASE XDDT
          SZL
          JMP TTSX         /XDDT. GIVE HIM R2.
GTYBSX,  SZF 5
          JMP TTHNG1       /A "DON'T HANG ME" IOT
          JSP HANGS        /HANG USER
          JSP ASC6         /REACTIVATE CHANNEL
          JMP HANG1

```

```

/SUBROUTINE TO HANG USER. HUNG STATUS IN I.O.
HANGS,   DAP HANGSX
          LAW I 7777       /LOOK AT TOP 6 BITS OF...
          LSM
          AND I ERSTAT     /...GUY'S STATUS.
          SWP
          SNI
          DIP I ERSTAT     /NO SPECIAL BITS SET; HANG HIM.
          ESM
HANGSX,  JMP .

```

```

/SUBROUTINE TO RE-ACTIVATE CH 6 AND ISB IF SCANNER STOPPED.
ASC6,    DAP ASC6X
          ASC 600
          CKS
          RIR 2S
          SPI
          ISB 600          /SCANNER STOPPED
ASC6X,   JMP .

```

/IOT'S TO LEAVE, ENTER, AND SAVE UP 8-BIT MODE AND CONTROL MODE.
TTMODE,

JSP TTCKS1
SAD C1
JMP TTERR /NOT HIS
LAC I TTFF
SZF 3
DAC 72 /RETURN OLD STATUS IN I.O.
LAW 4
SZF I 1 /8-BIT IOT?
JMP TTMOD1 /NO
LAC 70 /YES. SET NEW ALARM CHAR.
XOR I TTTP
AND C776000 /ALARM CHAR IN TOP 8 BITS
XOR I TTTP
LIO I TTTP
DAC I TTTP
SZF 3

TTMOD1,

DIO 70 /PUT OLD ALARM CHAR IN AC
LAW 20
DAC DTEM1 /STORE FLAG BIT FOR MODE
CMA
LSM
AND I TTTP
SZF 2 /ENTER MODE?
IOR DTEM1 /YES
DAC I TTTP

TTMODX,

ESM
JMP TTRETN

/IOT "GET TELETYPE"
GTY,

JSP TTSET
LIO RSTAT
LAW 377
LSM
AND I TTTP
SZA I
DIO I TTTP
ESM
SZA

/IF NO ONE'S: ASSIGN TO HIM

TTHNG1,

SAD RSTAT
JMP TTRETN
CLC
DAC 72
JSP ASC6
JMP TTRETN

/NORMAL RETURN

/"DON'T HANG ME" CODING

/=0 IN I.O. TO INDICATE HANG CONDITION

/REACTIVATE CHANNEL 6

/GIVE HIM A GOOD RETURN

/IOT RELEASE TT

RTY, JSP TTCKS1
 SAD (1)
 JMP TTERR
 DZM HOTFLG
 LAW 7741
 LSM
 AND I TTFF
 DAC I TTFF
 DZM I TTPP
 ESM
 JMP TTMODX

/NOT YOUR TT?

RTY1, DCH .

/IOT TO HANG USER UNTIL BREAK ON TT IN TTNO

DHANG, DSC 600
 JSP TTCKS1
 LIO (EOTHNG)
 SAS (2)
 JMP TTHANG
 JSP ASC6
 JMP DHANGX

/BREAK?

/NO. HANG HIM

/IN TTOK, CLEAR OUT BREAK AND UNHANG

/EXEC DDT TYO AND TYI

XDDTYO, JDA XDDTX

/CALLED FROM CORE 10

CLA

RCR 6S

JDA TTS

/TYPE FROM BOTTOM OF IO

JMP XDDTXR

XDDTXY,

LAI

LIO DDTFLG

LFI

LIA

JSP HNGDDT

XDDTX1,

CLL"U"CML

/XDDT FLAG FOR TT ROUTINES

LAC EXEC TT

JDA TTNUM

/SETUP POINTERS

LIO XDDTEM

XDDTXX,

JMP .

XDDTX,

0

/EXIT ACROSS CORE

DAP XDDTXX

DIO XDDTEM

LIF"U"SCF

DIO DDTFLG

JMP XDDTX1

XDDTXR,

LIO DDTFLG

/RETURN ACROSS CORE

LFI

LIO XDDTEM

JMP I XDDTX

XDDTYI,

JDA XDDTX

/CALLED FROM CORE 10

JSP TTU

/TYPE INTO TOP OF AC

JMP XDDTXR

DZM DDTGUY

JMP XDDTXY

XDDTEM,

0

EXEC TT,

0

DTEM1,

0

/ROUTINE FOR DISPATCHER TO HANG XDDT

```

HNGDDT,   DAC HDDTX
          LAW I 7777
          LSM
          AND STAT 0      /XDDT IS USER 0
          SWP
          SNI
          DIP STAT 0
          ESM
          JSP ASC6        /RE=ACTIVATE CH 6 AFTER HANGING DDT
HNGDD2,   LIF
          DIO DDTFLG
          JMP SWORG       /ON CH 17 BREAK = CALLED FROM XDDT

```

/ROUTINE TO START UP XDDT WHEN IT COMES INTO CORE

```

RUNDDT,   LAW I 7777
          LSM
          AND STAT 0
          SZA
          JMP DDTBRK      /BREAK KEY ON EXEC TT
RUNDD1,   ESM
          LAC RCORE
          DAC DCORE
          SAR 28
          IOR (200000)    /XDDT RUNS IN CORE 10 (RENAMED)
          DCH (ADD .+1)
          RNM
          LAC DDTGUY
          SZA             /DID HE HAVE ANYBODY
          JMP DDTWT1      /YES, GET HIM INTO CORE AGAIN
RUNDD2,   EEM
          LIO DDTFLG
          LFI
          JMP I HDDTX     /NO, GO BACK TO DDT'S CORE

```

```

HDDTX,    100000
DDTGUY,    0
DDTSU,    100000
DDTNUS,    100001
DDTFLG,    0

```

```

DDTBRK,   LAW 1
          DAC STAT 0
          LAC DDTSU
          DAC HDDTX
          DZM DDTGUY
          JMP RUNDD1

```

/ROUTINE FOR XDDT TO CALL TO GET A GUY INTO CORE
DDTWNT, DAC HDDTX

LEM
LAW STAT

AAI
DAC DDTGUY

DDTWT1, SAD BDSTAT
LAW FSTAT
DAP DDTWT3
LAW WHERE=STAT
ADD DDTGUY
DAP DDTWT2
LAW 777

LSM

DDTWT2, AND .
SAD (300 /IS THERE SUCH A GUY
JMP DDTHLT /NO

LAC (300000

DDTWT3, AND .
CLI "U"SWP

SNI
LAC I DDTWT3

IOR (10000)

DIP I DDTWT3 /SET BIT

LAW 777 /MASK QUEUE IN CASE GUY WAS CLKHNG

AND I DDTWT3

DAP I DDTWT3

ESM

LAW I 2

DAC I DCORE /HOLD DDT'S CORE

JMP HNGDD2

/ROUTINE TO START UP XDDT WHEN HIS USER COMES INTO CORE
DDTWS, LAC (=10000) /REMOVE XDDT WANTS BIT

LSM

AND I RSTAT

DIP I RSTAT

ESM

LAC RSTAT /IS THIS GUY DDT CURRENTLY WANTS

SAS DDTGUY

JMP SWORG /NOT GUY; FORGET IT

SAD SPOWN

DZM SPOWN /IN CASE DDT CHANGES HIS PC

LAW STAT 0 /IS GUY; SET DDT RUNNING

DAC I DCORE

DAC RSTAT

DAP ERSTAT

JMP RUNDD2

DDTHLT, ESM
LAC DDTNUS /TELL DDT THAT THERE'S NO USER

DAC HDDTX

JMP RUNDD2

/XDDT RPA

DDTRPA, DAC DDTRPX
DDTRP1, JSP RDA
XX
JMP DDTRDH
JMP I DDTRPX

DDTRPX, 0

DDTRDH, LIO (RPAHNG)
JSP HNGDDT
JMP DDTRP1

START